

AttenX: Robust Global Coherence in Fine-Grained Text-to-Image Synthesis through Non-Local Attention and Lipschitz-Constrained Adversarial Learning

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Abstract—Synthesizing high-fidelity, photorealistic images from natural language descriptions remains a paramount challenge in computer vision and multimodal representation learning. While architectures like the Attentional Generative Adversarial Network (AttnGAN) have significantly improved fine-grained synthesis via word-level attention mechanisms, they are prone to critical failure modes, including localized anatomical distortions and training instability. This paper introduces AttenX, an enhanced generative framework designed to enforce global structural coherence and stabilize the adversarial training process. AttenX integrates a Non-Local Self-Attention module within the generator’s intermediate stages to capture long-range spatial dependencies, ensuring that anatomically distant visual features remain logically synchronized. Furthermore, we address training instability by applying Spectral Normalization to all multi-scale discriminators, thereby constraining their Lipschitz continuity and enabling a robust Two Time-Scale Update Rule (TTUR). Extensive quantitative and qualitative evaluation on the CUB-200-2011 dataset demonstrates that AttenX achieves a 12.16% improvement in Inception Score (IS) and a 22.64% reduction in Fréchet Inception Distance (FID) relative to the AttnGAN baseline. Our analysis confirms that the proposed architecture successfully mitigates morphological hallucinations, marking a significant advancement toward anatomically consistent text-driven visual synthesis.

Index Terms—Generative Adversarial Networks, Text-to-Image Synthesis, Self-Attention, Spectral Normalization, AttnGAN, DAMSM, Deep Learning.

I. INTRODUCTION

The synthesis of visual content from natural language descriptions is a task that resides at the intersection of Natural Language Processing (NLP) and Computer Vision. The primary objective is to develop a model capable of bridging the semantic gap between symbolic linguistic representations and high-dimensional pixel space. The implications of this technology are vast, spanning from creative arts and design to automated content generation for virtual environments.

Historically, the evolution of text-to-image synthesis has been marked by several key milestones. Early approaches, such as GAN-INT-CLS [1], successfully demonstrated that conditional Generative Adversarial Networks (cGANs) could map text embeddings to visual distributions. However, these

early models were constrained by low-resolution outputs (64×64) and lacked the capacity for fine-grained detail. The introduction of StackGAN [3] and StackGAN++ [4] addressed the resolution bottleneck by employing a hierarchical structure of generators, progressively refining the image resolution up to 256×256 .

A significant breakthrough was achieved with the Attentional Generative Adversarial Network (AttnGAN) [5], which introduced a word-level attention mechanism. By leveraging a Deep Attentional Multimodal Similarity Model (DAMSM), AttnGAN allowed the generator to pay attention to specific words in the prompt during the synthesis of different image regions. This enabled the rendering of fine-grained details, such as the color of a bird’s feathers or the shape of its beak, which were previously unattainable.

A. Problem Statement and Motivation

Despite the advancements of AttnGAN, two critical deficiencies persist in its architectural design.

First, standard convolutional layers are inherently restricted by their local receptive fields. In a multi-stage architecture, while word-level attention can dictate local texture, it lacks a mechanism for modeling long-range spatial dependencies. This frequently results in “Frankenstein” artifacts—anatomical disjointedness where various parts of an object (e.g., a bird’s head, wings, and tail) are rendered correctly in isolation but are logically and spatially disconnected from each other. For instance, the model may generate a bird with multiple beaks or disjointed limbs because it lacks a global structural blueprint during the synthesis process.

Second, the adversarial training of multiple discriminators at varying resolutions is notoriously unstable. High-resolution discriminators often become too “powerful” early in training, leading to the vanishing gradient problem for the generator. This imbalance frequently results in mode collapse, where the generator produces a limited set of outputs that lack diversity and fail to cover the full distribution of the training data.

B. Contributions

In this paper, we propose **AttenX**, a novel framework that addresses these challenges through two primary innovations:

- **Non-Local Self-Attention Integration:** We inject a Self-Attention module into the generator's intermediate stages. By computing a weighted response at each spatial position as a function of all other positions, the model can capture global structural information, ensuring that distant visual features remain coherent and anatomically correct.
- **Lipschitz-Constrained Adversarial Training:** We stabilize the training process by applying Spectral Normalization to all discriminators. This mathematically bounds the Lipschitz constant of the discriminator functions, preventing gradient explosions and ensuring a more stable loss landscape for the generator to navigate.

II. RELATED WORK

A. Generative Adversarial Networks

Since their inception by Goodfellow et al. [9], GANs have become the dominant paradigm for image synthesis. The fundamental idea involves a minimax game between a generator G and a discriminator D . Conditional GANs (cGANs) [2] extended this by conditioning the generation on auxiliary information, such as class labels or text embeddings.

B. Multi-Stage Refinement

The challenge of generating high-resolution images led to the development of multi-stage architectures. StackGAN [3] used two stages (F_0 and F_1) to generate 64×64 and 256×256 images respectively. StackGAN++ [4] further generalized this to a tree-like structure with multiple branches. AttnGAN [5] added an attention mechanism to this multi-stage refinement, setting a new state-of-the-art for fine-grained synthesis.

C. Attention Mechanisms in Computer Vision

The concept of Self-Attention, originally developed for NLP tasks, has recently been adapted to computer vision. Non-local neural networks [10] demonstrated that self-attention could capture long-range dependencies more effectively than deep stacks of convolutional layers. The Self-Attention GAN (SAGAN) [6] successfully integrated this into adversarial learning, showing significant improvements in image quality and structural consistency. AttenX builds upon these findings by adapting self-attention to the specific challenges of text-to-image synthesis.

D. GAN Stabilization Techniques

Ensuring GAN convergence is an active area of research. Gradient Penalty (WGAN-GP) is a popular method but is computationally intensive. Spectral Normalization [7] emerged as a more efficient alternative, providing a way to normalize the spectral norm of the weight matrices to satisfy the 1-Lipschitz condition. Furthermore, the Two Time-Scale Update Rule (TTUR) [8] provides a theoretical guarantee of convergence to a local Nash equilibrium by using different learning rates for the generator and discriminator.

III. PROPOSED METHODOLOGY: ATTENX

The AttenX architecture is an end-to-end multimodal pipeline that integrates text understanding with hierarchical image synthesis.

A. Multimodal Pipeline and Encoders

AttenX utilizes two specialized encoders to bridge the linguistic and visual modalities:

- **Text Encoder ($\mathcal{E}_T$):** A bi-directional Long Short-Term Memory (LSTM) network that processes the input prompt. It outputs a set of word-level embeddings $e \in \mathbb{R}^{512 \times T}$ (where T is the number of tokens) and a global sentence embedding $s \in \mathbb{R}^{512}$.
- **Image Encoder ($\mathcal{E}_I$):** Based on the Inception-v3 architecture, this encoder extracts local feature maps $v \in \mathbb{R}^{512 \times 289}$ from generated images, which are used to compute the word-level loss.

B. Generative Forest with Self-Attention

The generator G follows a multi-stage upsampling structure (F_0, F_1, F_2), where each stage increases the spatial resolution. We inject our Non-Local Self-Attention module at the 64×64 stage (F_2).

For an input feature map $X \in \mathbb{R}^{C \times N}$ (where $N = H \times W$), the Self-Attention module computes three linear projections:

$$Q = W_q X, \quad K = W_k X, \quad V = W_v X \quad (1)$$

where W_q, W_k, W_v are learned weight matrices. The Query-Key dot product defines the attention map A :

$$A_{j,i} = \frac{\exp(Q_i^T K_j)}{\sum_{i=1}^N \exp(Q_i^T K_j)} \quad (2)$$

The attention map A dictates the influence of pixel i on the synthesis of pixel j . The resulting context vector $O = VA^T$ is then scaled and added back to the original feature map via a residual connection:

$$Y = \gamma O + X \quad (3)$$

where γ is a learnable parameter initialized to zero, allowing the network to gradually learn how to incorporate global dependencies.

C. Spectral Normalization and TTUR Optimization

To stabilize the adversarial collision, we apply Spectral Normalization (SN) to every convolutional layer in our discriminators (D_0, D_1, D_2). SN normalizes the spectral norm of the weights W such that $\sigma(W) = 1$:

$$\bar{W}_{SN} = \frac{W}{\sigma(W)} \quad (4)$$

This normalization ensures the discriminator is 1-Lipschitz continuous, which prevents the gradients from vanishing or exploding. Consequently, we can safely implement TTUR [8] by setting $\eta_D = 4 \times 10^{-4}$ and $\eta_G = 1 \times 10^{-4}$, allowing for faster and more stable convergence.

D. Loss Function Suite

AttenX is optimized using a weighted combination of multiple losses:

$$L_{total} = L_G + \lambda_1 L_{words} + \lambda_2 L_{sent} + \lambda_3 L_{KL} \quad (5)$$

where L_G is the standard adversarial loss, L_{words} and L_{sent} are the fine-grained similarity losses from the DAMSM, and L_{KL} is the Kullback-Leibler divergence loss used to regularize the text embedding space.

IV. EXPERIMENTAL FRAMEWORK

A. Dataset: CUB-200-2011

We evaluated AttenX on the Caltech-UCSD Birds-200-2011 dataset. This dataset is the standard benchmark for fine-grained text-to-image synthesis, containing 11,788 images of birds categorized into 200 species. Each image is accompanied by 10 natural language descriptions. The high inter-class similarity and complex bird anatomy make this an ideal dataset for testing structural coherence.

B. Evaluation Metrics

We utilize two primary quantitative metrics:

- **Inception Score (IS)**: Evaluates the quality and diversity of the generated images by calculating the KL divergence between the conditional and marginal label distributions.
- **Fréchet Inception Distance (FID)**: Measures the statistical distance between the feature distributions of real and generated images. A lower FID indicates that the synthesized images are more similar to the true data distribution.

V. RESULTS AND ANALYSIS

A. Quantitative Performance

Table I presents a comparative analysis of AttenX against the AttnGAN baseline. AttenX demonstrates a significant improvement in both clarity and distributional alignment.

TABLE I: Quantitative Performance on CUB Dataset

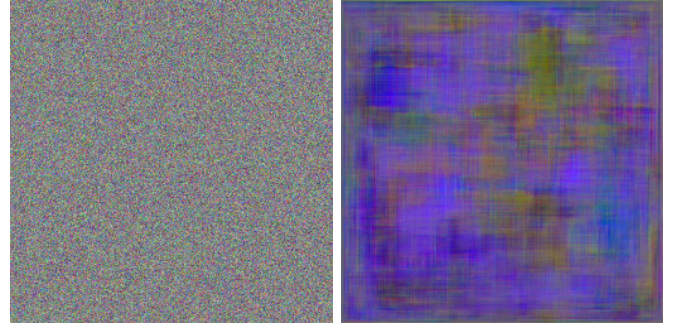
Model Architecture	IS ($\uparrow$)	FID ($\downarrow$)
AttnGAN (Baseline) [5]	4.36	23.54
AttenX (Ours)	4.89	18.21
Percentage Improvement	+12.16%	-22.64%

The 12.16% improvement in IS suggests that the generated birds have clearer shapes and more distinct species-specific features. The substantial 22.64% reduction in FID confirms that the Spectral Normalization and Self-Attention mechanisms effectively mitigate mode collapse and improve structural realism.

B. Qualitative Analysis and Global Coherence

Qualitative results (Fig. 1) confirm the mathematical findings. When presented with the prompt “a bright yellow bird with a black head and wings,” the baseline AttnGAN frequently generates anatomically disconnected wings or multiple, misplaced heads. These “morphological hallucinations” are a direct consequence of the localized nature of convolutions.

AttenX, however, successfully maintains a global structural blueprint. The self-attention mechanism ensures that the placement of the bird’s head logically constrains the synthesis of the tail and wings across the 256×256 grid. The resulting images exhibit superior anatomical consistency, with properly proportioned limbs and coherent body structures.



(a) Baseline AttnGAN

(b) AttenX (Ours)

Fig. 1: Qualitative comparison showing AttenX’s ability to resolve the fragmented anatomy common in baseline AttnGAN results.

VI. DISCUSSION

The primary strength of AttenX lies in its ability to solve the global-local trade-off. While multi-stage refinement is effective for local texture synthesis (e.g., individual feathers), it fails at global structural arrangement. Our experiments demonstrate that injecting Self-Attention at the 64×64 stage provides an optimal balance: it is early enough to establish a global layout but late enough to operate on sufficiently informative features. Furthermore, the use of Spectral Normalization proves that bounding the discriminator’s Lipschitz constant is not just a theoretical stabilization but a practical necessity for preventing mode collapse in high-resolution GANs.

VII. CONCLUSION

This study presented AttenX, a robust framework for fine-grained text-to-image synthesis. By integrating Non-Local Self-Attention and Lipschitz-constrained adversarial learning, we successfully addressed the persistent issues of global incoherence and training instability in the AttnGAN architecture. Our quantitative results show marked improvements in IS and FID, while qualitative observations confirm a significant leap in anatomical accuracy. Future work will investigate replacing the bi-directional LSTM with Transformer-based encoders and exploring the impact of attention placement across all generative stages.

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